

Seyfried metabolic diet — protein & fat

For caregivers and patients. Summary of the Seyfried / metabolic framing (adjunct to standard treatment).
Not a start-everything protocol.

Main line: Keep carbs and glucose very low. Healthy fats are the primary calorie source. Protein is moderate — enough to maintain muscle, not the dominant calorie source. Do not replace carbs with huge amounts of meat; replace most of those calories with healthy fat and protect muscle.

CARBS

No carbs or sugar unless it is an avocado or something that is mostly fat.

PROTEIN

- Whole food when tolerated: eggs, fish, clean meats — paired with healthy fats (avocado, olive oil, butter, ghee).
- **Bone broth** when eating is difficult. Add butter to bring fat up without more protein or carbs. Butter in unsweet tea works the same way.
- If a supplement is needed (muscle wasting / cachexia or cannot tolerate solid food): unsweetened bone broth protein (lower glutamine than whey/dairy) or clean unsweetened pea/rice protein — no added sugar, very low carbs. Use brands vetted with a clinician.
- Protein is a side, not a large plate of steak or chicken.

FAT

Fat is key. In the therapeutic frame, roughly 80–90% of calories from fat. Olive oil throughout the day on whatever the patient can eat; macadamia nut oil, MCT oil, avocado oil, ghee, butter.

GLUTAMINE CLARIFICATION

Moderate protein is **not** about eliminating glutamine through diet — the body can make glutamine itself. The goal is enough protein to preserve lean muscle while keeping glucose and GKI controlled, with fat as the main calorie source. If the patient is losing weight or muscle, reassess protein and calories upward — do not restrict harder to chase a number.

GKI

Glucose-Ketone Index: the relationship between blood glucose and ketones. Tracking both shows how the diet is affecting the patient — not just assuming a food is “keto.”

CLINICIAN

Exact protein and total calories should come from the treating team — weight, appetite, treatment, and whether muscle is being maintained.

Public education summary. Not medical advice. Decisions belong with the patient's clinicians. Research anchor: Mukherjee et al., *Communications Biology* 2019 (PMC6541653) — mouse model; diet adjunct only, not a cure claim.